Parcel100 - GeoJSON Construction Technical Design

Purpose: preserve a complete, implementation-level record of how Parcel100 constructs offline parcel GeoJSON files directly from an ArcGIS FeatureServer polygon layer. This document is written for GIS professionals, maintainers, and future reimplementation work.

Design objective

Parcel100 turns one deterministic 1 degree by 1 degree Master 4 geographic extent into as many as 100 independently loadable parcel GeoJSON files. The program queries only the parcel records that intersect each 0.1 degree cell, keeps user-selected attributes, converts Esri polygon geometry to standards-based GeoJSON, skips empty cells, and writes portable EPSG:4326 FeatureCollections.

Core invariants

- One Master 4 input always describes exactly one whole-degree west/east/south/north box.
- The same Master 4 always produces the same 100 numbered cell extents and filenames.
- The source is an ArcGIS FeatureServer polygon layer URL ending in a concrete layer number such as /FeatureServer/0.
- GeoJSON output is WGS 84 longitude/latitude (EPSG:4326).
- Existing completed files are preserved and skipped on resumed runs.

1. Inputs, validation, and Master 4 parsing

User inputs

- Master 4: west,east,south,north, for example 82w,81w,30n,31n.
- ArcGIS FeatureServer polygon layer URL.
- Fields to keep in each GeoJSON.
- Output folder and coordinate precision.

Validation

Each coordinate token is parsed as an absolute numeric value plus N, S, E, or W. West and south become negative. Longitude is constrained to +/-180 degrees and latitude to +/-90 degrees. West must be less than east; south must be less than north. Both spans must be exactly 1 degree and all four edges must fall on whole-degree boundaries.

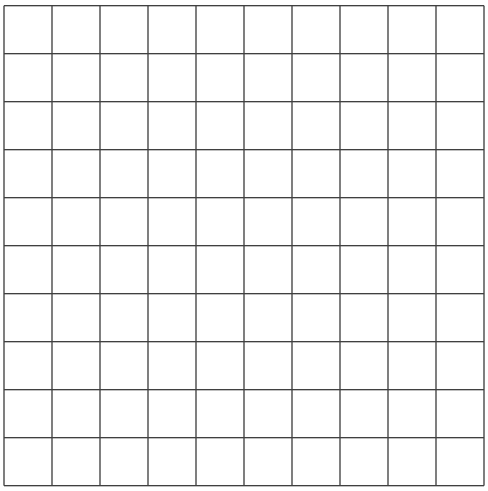
```
west = -82.0
east = -81.0
south = 30.0
north = 31.0
```

Canonical label

The parsed input is normalized into a deterministic label used later for filenames. Hemisphere prefixes and zero-padded degrees remove ambiguity. Example output prefix: W082W081N031N030.

2. Deterministic 10 x 10 cell construction

The one-degree Master 4 is subdivided into 100 equal geographic cells. Every cell is 0.1 degree wide by 0.1 degree high. Cell IDs run 001 through 100 using the established Master 4 Grid Maker numbering scheme.



Master 4 = 1 degree x 1 degree

100 cells = 0.1 degree x 0.1 degree

Cell IDs are deterministic: 001-100

Why the split matters

Parcel services can contain millions of polygons. Querying a whole Master 4 at once can produce huge responses, exceed service limits, trigger gateway timeouts, or create GeoJSON files that are slow to load on field hardware. The 100-cell architecture turns one massive extraction into predictable, independently usable pieces.

Seam behavior

Each cell query uses an ArcGIS envelope with spatial relationship esriSpatialRelIntersects. A parcel touching a cell boundary can therefore be returned by adjacent cells. That duplication is deliberate at file boundaries: it avoids visual gaps when neighboring cell files are loaded independently.

3. FeatureServer introspection and field selection

Read Service

Parcel100 first POSTs f=json to the concrete FeatureServer layer URL. The returned metadata is treated as the contract for the extraction. The program verifies that the service reports esriGeometryPolygon, reads the field definitions, layer name, object-ID field, and maxRecordCount.

```
POST <layer-url>
  f=json
```

Field selection

The GUI displays service fields using each field name plus alias when present. The output is intentionally attribute-selective: only fields selected by the user are copied into GeoJSON properties. This can reduce file size substantially compared with exporting the full tax-roll schema.

- Typical useful Florida fields: PARCEL_ID, OWN_NAME, PHY_ADDR1, PHY_CITY, PHY_ZIPCD.
- The recommended-field routine prefers known Florida names and otherwise falls back to generic owner/address/parcel/city/ZIP keywords.
- Geometry is never taken from the field list; it is requested separately as feature geometry.

4. Per-cell Object ID discovery

For each cell, Parcel100 first asks for object IDs only. This keeps the discovery response small and bypasses feature-count limits that affect ordinary feature queries.

```
POST <layer-url>/query
  f=json
  where=1=1
  geometry=<west>,<south>,<east>,<north>
  geometryType=esriGeometryEnvelope
  inSR=4326
  spatialRel=esriSpatialRelIntersects
  returnIdsOnly=true
  returnGeometry=false
```

Empty cells

If the response contains no objectIds, the cell is marked EMPTY and no file is written. This is important for coastal Master 4s and state-border grids where many of the 100 cells may contain ocean or territory outside the source dataset.

Four-part fallback

Version 1.3 hardens this stage. After repeated transient ID-query failures such as HTTP 504 Gateway Timeout, the same 0.1 degree cell is divided internally into four smaller envelopes. IDs are collected from all four queries, de-duplicated, and then processed normally. The external file architecture remains one file for the original cell.

5. Feature download batching and coordinate system

Once object IDs are known, Parcel100 retrieves complete polygon features in batches. The batch size is capped by the layer maxRecordCount and by an internal ceiling of 2,000 IDs.

```
POST <layer-url>/query
  f=json
  objectIds=101,102,103,...
  outFields=<selected fields>
  returnGeometry=true
  outSR=4326
  returnZ=false
  returnM=false
  geometryPrecision=<user precision>
```

Why outSR=4326

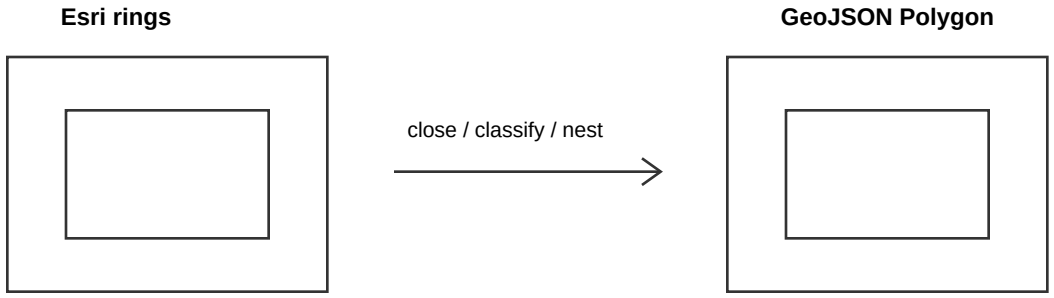
The source layer may use another coordinate reference system. The Florida Statewide Cadastral service, for example, reports NAD83(2011) / Florida GDL Albers (EPSG:6439). Parcel100 asks the ArcGIS service to return output geometry in WGS 84 (EPSG:4326), the coordinate system expected by conventional GeoJSON consumers and handled cleanly by ArcGIS Earth.

Retries

Network and service failures are retried with increasing short delays. Version 1.3 also reports retries visibly in the GUI so a slow public service does not appear to be a frozen application.

6. Esri rings to GeoJSON Polygon / MultiPolygon

ArcGIS polygon geometry is returned as one or more rings. GeoJSON requires explicit Polygon or MultiPolygon coordinate nesting. Parcel100 therefore reconstructs topology instead of blindly copying ring arrays.



Conversion sequence

- Discard malformed points and reduce each point to X/Y; Z and M are not requested.
- Close each ring if its last coordinate does not already equal its first coordinate.
- Discard degenerate zero-area rings.
- Compute signed area magnitude for every ring.
- Build a containment tree: for each ring, find the smallest larger ring containing a representative point.
- Even containment depth becomes an outer polygon; its immediate odd-depth children become holes.
- One reconstructed outer becomes GeoJSON Polygon. Multiple outers become MultiPolygon.

This containment approach is intentionally independent of winding direction because real-world services do not always preserve Esri ring orientation consistently after processing or reprojection.

7. GeoJSON Feature and FeatureCollection assembly

Per-feature structure

Each ArcGIS feature becomes one GeoJSON Feature. Properties are rebuilt from only the selected field names. Geometry is the converted Polygon/MultiPolygon object, or null only if the source returned no geometry.

```
{
  "type": "Feature",
  "properties": {"PARCEL_ID": "...", "OWN_NAME": "..."},
  "geometry": {"type": "Polygon", "coordinates": [...] }
}
```

Per-cell collection

All features for one 0.1 degree cell are written as a compact GeoJSON FeatureCollection. Parcel100 also records useful top-level provenance: the cell bbox, original Master 4 label, numeric cell ID, and source FeatureServer URL.

```
{
  "type": "FeatureCollection",
  "name": "W082W081N031N030-006-Parcels",
  "bbox": [west,south,east,north],
  "master4": "82W_81W_30N_31N",
  "cell": 6,
  "source": "https://.../FeatureServer/0",
  "features": [...]
}
```

8. File naming, resume behavior, and data integrity

Filename convention

The filename itself is geographic metadata. Example: W082W081N031N030-006-Parcels.geojson. It identifies the Master 4 and the 001-100 cell without opening the file.

Resume behavior

Before making any server request for a cell, Parcel100 computes the exact expected output filename and checks the output folder. If that file already exists, the cell is treated as completed and skipped. Existing files are never overwritten automatically.

Why this matters

Long public-service extractions can be interrupted by 504 responses, temporary disconnects, sleep events, or application closure. Resume semantics make a partial output folder durable work rather than a failed run. Restarting against the same Master 4 and folder continues at the first missing file.

- Empty cells generate no placeholder GeoJSON.
- Completed files remain untouched on a resumed build.
- The progress bar advances across all 100 logical cells, including existing and empty cells.
- Counts distinguish newly written, existing skipped, empty skipped, and parcel records written during the current run.

9. Operational validation and service behavior

Florida test sources

Two public ArcGIS-hosted Florida parcel services were exercised with the same Parcel100 workflow. The digitalRE FL_Parcels FeatureServer on services5 behaved as the practical high-throughput source. The Florida Statewide Cadastral FeatureServer on services9 provided official-source lineage but repeatedly exhibited slow responses, 504 Gateway Timeout errors, and occasional generic query failures.

```
services5: https://services5.arcgis.com/GcvM6vDlR2gM4x31/arcgis/rest/services/FL_Parcels/FeatureServer/0
services9: https://services9.arcgis.com/Gh9awoU677aKree0/arcgis/rest/services/Florida_Statewide_Cadastral/Feat
```

Observed production runs

- Grid 3 (83W,82W,30N,31N), services5: 63 GeoJSON files, 37 empty cells, 78,837 parcel records; about two minutes for the full 100-cell pass.
- Grid 4, services5: 99 GeoJSON files, 1 empty cell, 469,146 parcel records.
- Earlier M2 run, services5: 92 files, 8 empty cells, 547,183 records; approximately 314 MB total.
- services9 cell 022 required the four-part ID fallback and successfully recovered 11,969 parcel IDs; later 504 download retries also recovered.

Interpretation: the data model and Parcel100 pipeline were not the bottleneck. Identical application logic ran dramatically faster against services5, isolating the operational instability to the services9 delivery path.

10. End-to-end pseudocode

```
read layer metadata
validate polygon geometry
read fields and maxRecordCount
parse Master 4
generate cells 001..100

for each cell:
    path = expected output filename
    if path exists:
        skip as completed
        continue

    ids = query IDs intersecting cell envelope
    if transient failure repeats:
        ids = union of four sub-envelope ID queries

    if ids is empty:
        mark EMPTY
        continue

    features = []
    for each ID batch:
        response = query selected fields + geometry, outSR=4326
        for each Esri feature:
            geometry = rebuild rings as Polygon/MultiPolygon
            properties = selected attributes only
            append GeoJSON Feature

    write compact FeatureCollection atomically to cell filename
```

Reimplementation warning

Do not replace the two-stage ID-then-feature workflow with one giant feature query. The separation is what makes the extraction compatible with large statewide services and maxRecordCount limits. Likewise, do not remove deterministic cell numbering or resume-by-filename without replacing them with an equally explicit durability mechanism.

11. Maintainer checklist / future implementation notes

Must preserve

- Master 4 whole-degree validation and established 001-100 numbering.
- FeatureServer layer validation (polygon geometry).
- ID-only spatial discovery before feature retrieval.
- Explicit outSR=4326 for generated GeoJSON.
- Selective attributes, not blind full-schema export.
- Robust ring closure and containment-based Polygon/MultiPolygon reconstruction.
- Empty-cell skipping and deterministic filenames.
- Existing-file resume semantics.
- Visible retry status and bounded retry behavior.
- Four-part fallback for repeatedly failing cell ID queries.

Safe future extensions

- Adaptive feature-download batch reduction after repeated 504 responses.
- Service presets by state while retaining arbitrary FeatureServer support.
- Optional per-cell manifest containing source timestamp, field list, record count, and checksum.
- Automated validation that generated GeoJSON loads and identifies correctly in ArcGIS Earth or another standards-compliant viewer.

Design summary

Parcel100 is intentionally not a GIS desktop replacement. It is a deterministic extraction and translation engine: geographic cell -> spatial IDs -> selected ArcGIS features -> normalized WGS84 geometry -> portable GeoJSON. The value is the repeatability, manageable file size, offline portability, and the removal of a multi-step expert GIS workflow from the field-map production process.